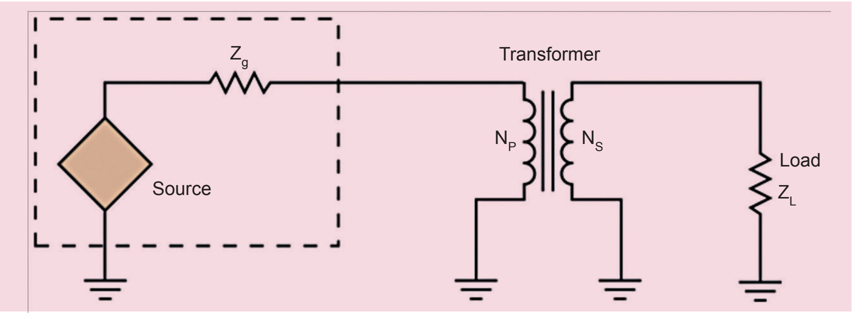

Fig. 2: Transformer based impedance matching

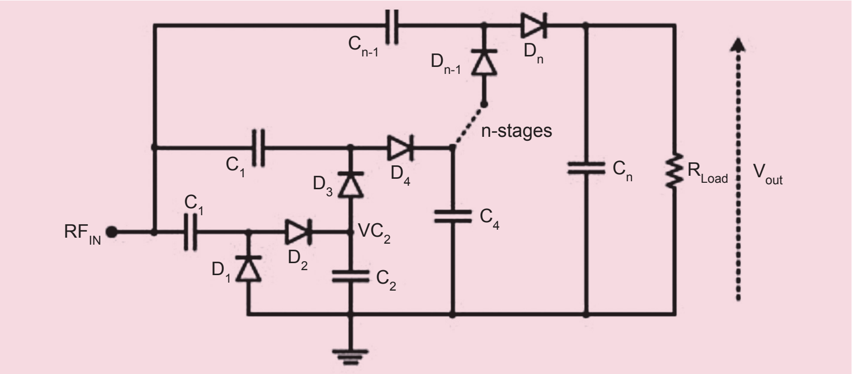

Fig. 3: n-stage Dickson voltage rectifier

Let us consider an example of RF energy harvesting system, where, to increase the conversion efficiency, the designed circuit should resonate at the targeted frequency. A matching circuit has to be designed to match the impedance of the multiplier circuit to the standard 50-ohm antenna for allowing maximum power transfer from the antenna to the circuit.

To match the impedance, it is necessary to first identify the input impedance of the rectifier circuit, which generally consists of nonlinear devices such as diodes and transistors. Small-signal modeling techniques for approximating the behaviour of these nonlinear devices cannot be used due to the large input signal and absence of an input DC bias. The large input signals without any DC bias results in the rectifying device to operate in different regions of operation within the same input cycle. This leads to variation in the input impedance of the rectifier circuit.

For such systems, designing impedance matching circuits is complicated. You need to identify input impedance under all the operating regions such as the subthreshold region, ohmic region, and saturated region according to different DC-bias points, and then design multiple matching circuits accordingly.

However, small variation in input signal guarantees that small-signal parameters of the transistors remain relatively constant, and the calculated input impedance accurately predicts the behavior of the circuit during the normal operation. A fixed impedance matching circuit can then be designed.

There are two popular methods by which impedance matching can be executed, one by fixed LC (inductor-capacitor) impedance matching circuits, and the other by using a transformer to achieve impedance matching. Transformer based matching has an advantage of lower die area and robustness compared to the LC matching method.

Let us first have a look at the LC based matching method. We start with simple topology consisting of an inductor and a capacitor. The difference between the impedances of the rectifier and the transducer dictates the quality factor and the values of capacitor and inductor.

There are four different possible versions of LC circuit, two of them would be low-pass and the other two would be high-pass versions. The low-pass versions are generally employed in RF (radio frequency) circuits since they attenuate harmonics, noise, and other undesired signals. But in our case, we need energy sources from the maximum bandwidth possible. Therefore, the topology shown in Fig. 1 works best in our case.

The circuit in Fig. 1 works when the input impedance of the rectifier is greater than the output impedance of the transducer. For identifying the values of inductor and capacitor, you need both the aforementioned impedances and a desired frequency. Then you need to plugin the numbers in the following set of equations:

$$X_L = \sqrt{(R_S \times (R_L - R_S))}$$
$$X_C = (R_S \times R_L)/X_L$$

where R_S is the output resistance of the transducer and R_L is the input impedance of the rectifier circuit. X_L and X_C are the reactances from which you can get the values of L and C.

The quality factor of these circuits can only be changed by changing the values of L and C. The quality factor determines the steepness of the frequency response. Low quality factor will be very broadband and will not filter out undesired frequencies, which is desired in our case. There are many other versions of impedance matching circuits like T or pi-network where we can control the quality factor, but in this case low quality factor is desirable and can be easily achieved.

Next, let us take a look at transformer based impedance matching. This technique is preferred when robustness is required. Moreover, a transformer offers a close ideal method for making one impedance look like another. It does this by using the turns ratio (N), which is N_S (turns on the secondary side of the transformer) divided by N_P (turn on the primary side of the transformer).

The relationship of the impedances can be calculated as:

$$Z_S \div Z_P = (N_S \div N_P)2$$

where Z_P represents the output impedance of the transducer and Z_S represents the input impedance of the rectifier circuit.

Most of the energy harvesting circuits use an inductor based matching circuit as it is easier to implement on-chip, but transformer based matching